Prayas JEE AIR (2025)

WEEKLY TEST 25

DURATION : 180 Minutes | **DATE : 29/12/2024** | **M.MARKS : 300**

Topics Covered

Physics:	Full Syllabus
Chemistry:	Full Syllabus
Mathematics:	Full Syllabus

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration.
3. The test booklet consists of 75 questions. The maximum marks are **300**.
4. There are three Sections in the question paper, Section I, II & III consisting of Section-I **(Physics)**, Section-II **(Chemistry)**, Section-III **(Mathematics)** and having **25** questions in each part in which first **20** questions are of Objective Type and Last **5** questions are integers type and all **25** questions are **compulsory.**
5. There is only one correct response for each question.
6. Each correct answer will give **4** marks while **1** Mark will be deducted for a wrong response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.
9. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number: _____

OMR Bar Code Number: _____

Candidate's Signature: _____ **Invigilator's Signature** _____

SECTION-I (PHYSICS)

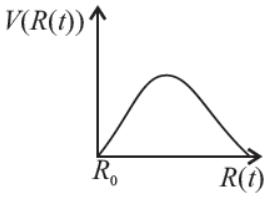
Single Correct Type Questions

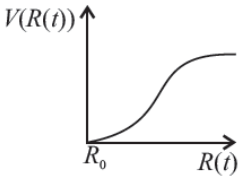
1. A juggler throws ball vertically upwards with same initial velocity in air. When the first ball reaches its highest positions, he throws the next ball. Assuming the juggler throws n balls per second, the maximum height the balls can reach is

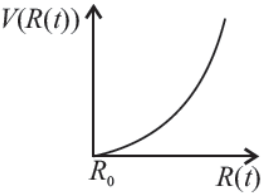
(1) $g/3n^2$ (2) $2\,gn$

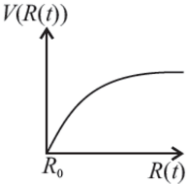
(3) g/n^2 (4) $g/2n^2$

2. There is a uniform spherically symmetric surface charge density at a distance R_0 from the origin. The charge distribution is initially at rest and starts expanding because of mutual repulsion. The figure that represents best the speed $V(R(t))$ of the distribution as a function of its instantaneous radius $R(t)$ is

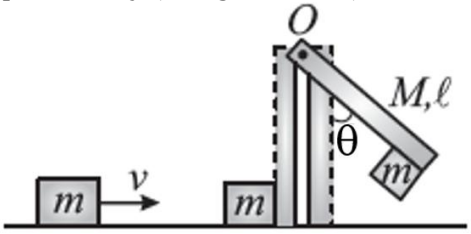
(1) 

(2) 

(3) 

(4) 

3. A block of mass m = 1 kg slides with velocity v = 6 m/s on a frictionless horizontal surface and collides with a uniform vertical rod and sticks to it as shown. The rod is pivoted about O and swings as a result of the collision making angle θ before momentarily coming to rest. If the rod has mass M = 2 kg and length l = 1 m, the value of θ is approximately (take g = 10 m/s^2)



(1) 49° (2) 55°

(3) 63° (4) 79°

4. A current of 1.5 A is flowing through a triangle, of side 9 cm each. The magnetic field at the centroid of the triangle is (Assume that the current is flowing in the clockwise direction.):

(1) $2\sqrt{3}\times10^{-7}$ T, outside the plane of triangle

(2) $2\sqrt{3}\times10^{-5}$ T, inside the plane of triangle

(3) 3×10^{-5} T, inside the plane of triangle

(4) 3×10^{-7} T, outside the plane of triangle

5. Match the list:-

List-1		List-II	
A	Moment of inertia of the rod (length L, Mass M, about an axis ⊥ to the rod passing through the midpoint	P	$\frac{8ML^2}{3}$
B	Moment of inertia of the rod (length L, Mass $2M$, about an axis ⊥ to the rod Passing through one of its end)	Q	$\frac{ML^2}{3}$
C	Moment of inertia of the rod (length $2L$, Mass M, about an axis ⊥ to the rod Passing through its midpoint)	R	$\frac{ML^2}{12}$
D	Moment of inertia of the rod (length $2L$, Mass $2M$, about an axis ⊥ to the rod Passing through one of its end)	S	$\frac{2ML^2}{3}$

Choose the correct answer from the options given below:

(1) A → Q; B → R; C → S; D → P

(2) A → R; B → S; C → P; D → Q

(3) A → R; B → S; C → Q; D → P

(4) A → Q; B → R; C → P; D → S

6. A vessel of depth 'd' is half filled with oil of refractive index n_1 and the other half is filled with water of refractive index n_2. The apparent depth of this vessel when viewed from above will be

(1) $\frac{dn_1n_2}{(n_1+n_2)}$

(2) $\frac{d(n_1+n_2)}{2n_1n_2}$

(3) $\frac{dn_1n_2}{2(n_1+n_2)}$

(4) $\frac{2d(n_1+n_2)}{n_1n_2}$

7. The area of cross section of the rope used to lift a load by a crane is 2.5×10^{-4} m^2. The maximum lifting capacity of the crane is 10 metric tons. To increase the lifting capacity of the crane to 25 metric tons, the required area of cross section of the rope should be: (take g = 10 ms^{-2})

(1) 6.25×10^{-4} m^2 (2) 10×10^{-4} m^2

(3) 1×10^{-4} m^2 (4) 1.67×10^{-4} m^2

8. A plane electromagnetic wave is propagating along the direction $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$, with its polarization along the direction $\hat{k}$. The correct form of the magnetic field of wave would be (here B_0 is an appropriate constant and K is angular wave number)

(1) $B_0 \frac{\hat{i}+\hat{j}}{\sqrt{2}} \cos\left(\omega t - K\frac{r\hat{i}+r\hat{j}}{\sqrt{2}}\right)$

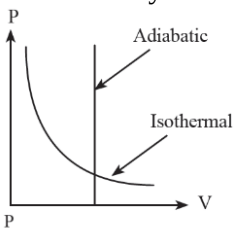
(2) $B_0 \frac{\hat{i}-\hat{j}}{\sqrt{2}} \cos\left(\omega t + K\frac{r\hat{i}+r\hat{j}}{\sqrt{2}}\right)$

(3) $B_0 \frac{\hat{i}-\hat{j}}{\sqrt{2}} \cos\left(\omega t - K\frac{r\hat{i}+r\hat{j}}{\sqrt{2}}\right)$

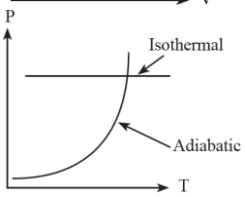
(4) $B_0 \hat{k} \cos\left(\omega t - K\frac{r\hat{i}+r\hat{j}}{\sqrt{2}}\right)$

9. Which one is the correct option for the two different thermodynamic processes?

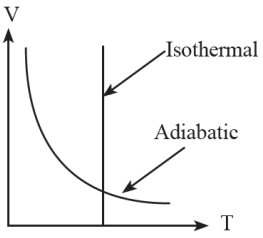
(A)



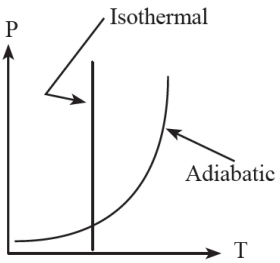
(B)



(C)



(D)



(1) (C) and (A) (2) (B) and (C)

(3) (C) and (D) (4) (A) only

10. You are given that Mass of ${}^{7}_{3}Li = 7.0160$ amu Mass of ${}^{4}_{2}He = 4.0026$ amu and ${}^{1}_{1}H = 1.0079$ amu When 20 g of ${}^{7}_{3}Li$ is converted into ${}^{4}_{2}He$ by proton capture, the energy liberated, (in kWh), is [Mass of nucleon = $931 \frac{\text{MeV}}{c^2}$]

(1) 4.5×10^5 (2) 8×10^6

(3) 6.82×10^5 (4) 1.33×10^6

11. Given below are two statements:

Statement-I: An electric dipole is placed at the center of a hollow sphere. The flux of electric field through the sphere is zero but the electric field is not zero anywhere in the sphere.

Statement-II: If R is the radius of a solid metallic sphere and Q be the total charge on it. The electric field at any point on the spherical surface of radius $r(<R)$ is zero but the electric flux passing through this closed spherical surface of radius r is not zero.

In the light of the above statements. Choose the correct answer from the option given below:

(1) Statement-I is true but Statement-II is false

(2) Statement-I is false but Statement-II is true

(3) Both Statement-I and Statement-II are true

(4) Both Statement-I and Statement-II are false

12. A particle is projected with velocity v_0 along x-axis. A damping force is acting on the particle which is proportional to the square of the distance from the origin i.e. $ma = -\alpha x^2$. The distance at which the particle stops:

(1) $\left(\frac{2v_0}{3\alpha}\right)^{\frac{1}{3}}$ (2) $\left(\frac{3v_0^2}{2\alpha}\right)^{\frac{1}{2}}$

(3) $\left(\frac{2v_0^2}{3\alpha}\right)^{\frac{1}{2}}$ (4) $\left(\frac{3v_0^2 m}{2\alpha}\right)^{\frac{1}{3}}$

13. A moving coil galvanometer, having a resistance G, produces full scale deflection when a current I_g flows through it. This galvanometer can be converted into (i) an ammeter of range 0 to I_0 ($I_0 > I_g$) by connecting a shunt resistance R_A to it and (ii) into a voltmeter of range 0 to $V(V = GI_0)$ by connecting a series resistance R_V to it. Then,

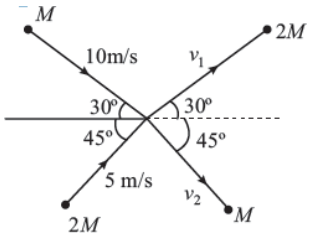
(1) $R_A R_V = G^2\left(\frac{I_g}{I_0 - I_g}\right)$ and $\frac{R_A}{R_V} = \left(\frac{I_0 - I_g}{I_g}\right)^2$

(2) $R_A R_V = G^2$ and $\frac{R_A}{R_V} = \left(\frac{I_g}{I_0 - I_g}\right)^2$

(3) $R_A R_V = G^2$ and $\frac{R_A}{R_V} = \frac{I_g}{I_0 - I_g}$

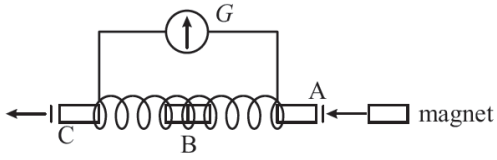
(4) $R_A R_V = G^2\left(\frac{I_0 - I_g}{I_g}\right)$ and $\frac{R_A}{R_V} = \left(\frac{I_g}{I_0 - I_g}\right)^2$

14. Two-particles, of masses M and $2M$, moving as shown, with speeds of 10 m/s and 5 m/s collide elastically at the origin. After the collision, they move along the indicated directions with speeds v_1 and v_2 respectively. The value of v_1 and v_2 are nearly:

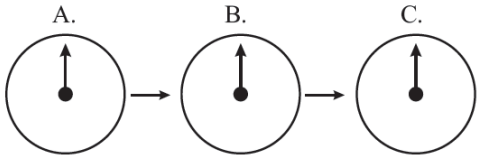


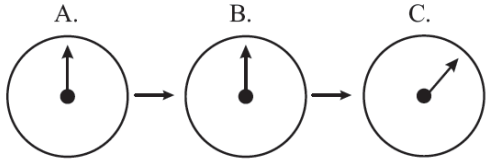
(1) 3.2 m/s and 6.3 m/s
(2) 3.2 m/s and 12.6 m/s
(3) 6.5 m/s and 6.3 m/s
(4) 6.5 m/s and 3.2 m/s

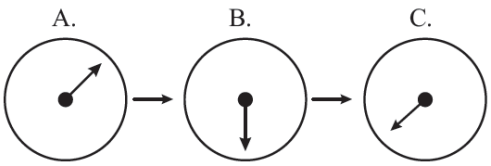
15. A small bar magnet is moved through a coil at constant speed from one end to the other. Which of the following series of observations will be seen on the galvanometer G attached across the coil?

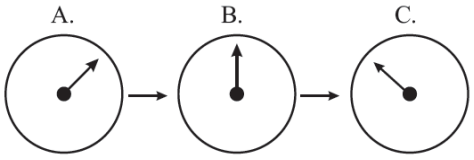


Three positions shown describe:
A. the magnet's entry
B. magnet is completely inside and
C. magnet's exit.

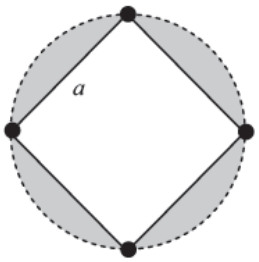
(1) 

(2) 

(3) 

(4) 

16. Four identical particles of mass M are located at the corners of a square of side 'a'. What should be their speed if each of them revolves under the influence of other's gravitational field in a circular orbit circumscribing the square?



(1) $1.21\sqrt{\frac{GM}{a}}$

(2) $1.41\sqrt{\frac{GM}{a}}$

(3) $1.16\sqrt{\frac{GM}{a}}$

(4) $1.35\sqrt{\frac{GM}{a}}$

17. The width of fringe is 2 mm on the screen in a double slit experiment for the light of wavelength of 400 nm. The width of the fringe for the light of wavelength 600 nm will be:
(1) 4 mm (2) 1.33 mm
(3) 3 mm (4) 2 mm

18. A fully loaded aircraft has a mass of 5.4×10^5 kg. Its total wing area is 500 m^2. It is in level flight with a speed of 1080 km/h. If the density of air is 1.2 kg m^{-3}, the fractional increase in the speed of the air on the upper surface of the wings relative to the lower surface in percentage will be
($g = 10$ m/s^2)
(1) 16 (2) 6
(3) 8 (4) 10

19. A free electron of 2.6 eV energy collides with a H^+ ion. This results in the formation of a hydrogen atom in the first excited state and a photon is released. Find the frequency of the emitted photon. ($h = 6.6 \times 10^{-34}$ J-s)
(1) 0.19×10^{15} MHz (2) 1.45×10^9 MHz
(3) 1.45×10^{16} MHz (4) 9.0×10^{27} MHz

20. A particle is making simple harmonic motion along the x-axis. If at a distances x_1 and x_2 from the mean position the velocities of the particle are v_1 and v_2 respectively. The time period of its oscillation is given as:

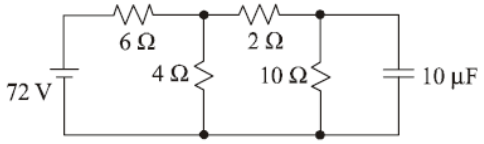
(1) $T = 2\pi\sqrt{\frac{x_2^2 - x_1^2}{v_1^2 - v_2^2}}$ (2) $T = 2\pi\sqrt{\frac{x_2^2 + x_1^2}{v_1^2 - v_2^2}}$

(3) $T = 2\pi\sqrt{\frac{x_2^2 - x_1^2}{v_1^2 + v_2^2}}$ (4) $T = 2\pi\sqrt{\frac{x_2^2 + x_1^2}{v_1^2 + v_2^2}}$

Integer Type Questions

21. A wire of density 9×10^{-3} kg cm^{-3} is stretched between two clamps 1 m apart. The resulting strain in the wire is 4.9×10^{-4}. The lowest frequency (in Hz) of the transverse vibrations in the wire is (Young's modulus of wire Y = 9×10^{10} Nm^{-2}), (to the nearest integer), _____.

22. Determine the charge (in μC) on the capacitor in the following circuit:



23. A 60 Hz AC voltage of 200 V impressed across an *LR*-circuit results in a current of 2 A. If the power dissipation is 320 W, calculate the rms value of the back emf (in volt) arising in the inductance.

24. In a photoelectric experiment, the wavelength of the light incident on a metal is changed from 300 nm to 400 nm. The decrease in the stopping potential (in V) is $\left(\frac{hc}{e} = 1240\,\text{nm} - \text{V}\right)$ Write answer to its nearest integer.

25. A bakelite beaker has volume capacity of 500 cc at 30°C. When it is partially filled with V_m volume (at 30°C) of mercury, it is found that the unfilled volume of the beaker remains constant as temperature is varied. If $\gamma_{(\text{beaker})} = 6 \times 10^{-6}\ °\text{C}^{-1}$ and $\gamma_{(\text{mercury})} = 1.5 \times 10^{-4}\ °\text{C}^{-1}$, where γ is the coefficient of volume expansion, then V_m (in cc) is close to _____.

SECTION-II (CHEMISTRY)

Single Correct Type Questions

26. The compound that cannot act both as oxidising and reducing agent is:

(1) HNO_2 (2) H_3PO_4

(3) H_2SO_3 (4) H_2O_2

27. Which of the following are isostructural pairs ?

A. SO_4^{2-} and CrO_4^{2-}

B. $SiCl_4$ and $TiCl_4$

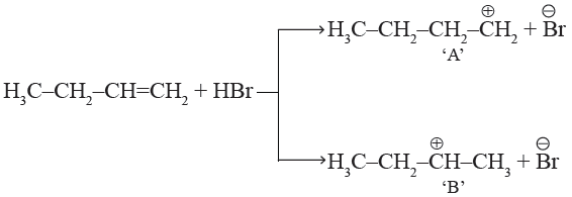
C. NH_3 and NO_3^-

D. BCl_3 and $BrCl_3$

(1) C and D only (2) A and B only

(3) A and C only (4) B and C only

28. Choose the correct statement regarding the formation of carbocations A and B given.



(1) Carbocation A is more stable and formed relatively at slow rate

(2) Carbocation B is more stable and formed relatively at faster rate

(3) Carbocation B is more stable and formed relatively at slow rate

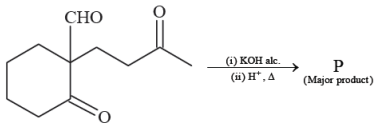
(4) Carbocation A is more stable and formed relatively at faster rate

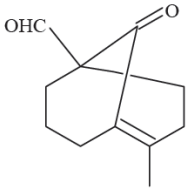
29. Given the equilibrium constant K_c of the reaction: $Cu(s) + 2Ag^+(aq) \rightarrow Cu^{2+}(aq) + 2Ag(s)$ is 10×10^{15}, Calculate the E°_{cell} of this reaction at 298 K $\left[2.303\frac{RT}{F} \text{ at } 298\,\text{K} = 0.059\,\text{V}\right]$

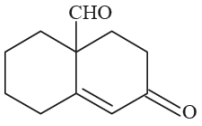
(1) 0.4736 mV (2) 0.4736 V

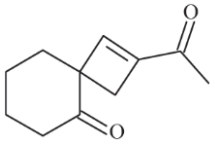
(3) 0.04736 V (4) 0.04736 Mv

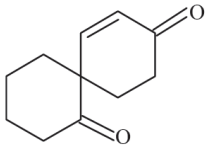
30. The major product (P) in the following reaction is:



(1) 

(2) 

(3) 

(4) 

31. Given below are two statements : One is labelled as Assertion (A) and the other is labelled as Reason (R):

Assertion (A): Zero orbital overlap is an out of phase overlap.

Reason (R): It results due to different orientation / direction of approach of orbitals.

In the light of the above statements, choose the correct answer from the options given below

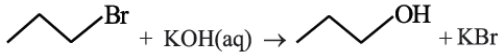
(1) Both A and R are true and R is the correct explanation of A

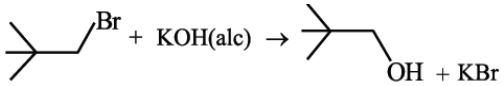
(2) Both A are R are true but R is NOT the correct explanation of A

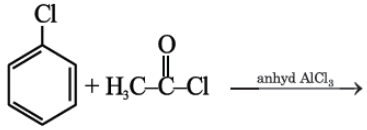
(3) A is true but R is false

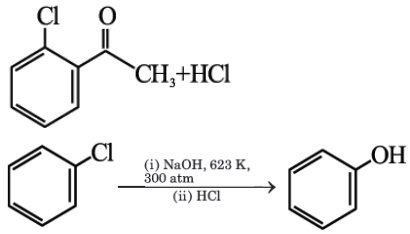
(4) A is false but R is true

32. Identify the incorrect option from the following:

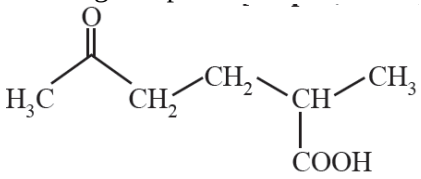
(1) 

(2) 

(3) 

(4) 

33. The correct IUPAC nomenclature for the following compound is



(1) 5–Formyl–2–methylhexanoic acid

(2) 2–Methyl–5–oxohexanoic acid

(3) 2–Formyl–5–methylhexan–6–oic acid

(4) 5–Methyl–2–oxohexan–6–oic acid

34. Liquids A and B form an ideal solution in the entire composition range. At 350 K, the vapor pressures of pure A and pure B are 7×10^3 Pa and 12×10^3 Pa, respectively. The composition of the vapour in equilibrium with a solution containing 40 mole percent of A at this temperature is

(1) $Y_A = 0.76$; $Y_B = 0.24$

(2) $Y_A = 0.37$; $Y_B = 0.63$

(3) $Y_A = 0.28$; $Y_B = 0.72$

(4) $Y_A = 0.4$; $Y_B = 0.6$

35. Match Column-I with Column-II.

Column-I Element detected		**Column-II Reagent used/ Product formed**	
A	Nitrogen	P	$Na_2[Fe(CN)_5 NO]$
B	Sulphur	Q	$AgNO_3$
C	Phosphorous	R	$Fe_4[Fe(CN)_6]_3$
D	Halogen	S	$(NH_4)_2 MoO_4$

Choose the correct answer from the options given below:

(1) A→Q; B→S; C→P; D→R

(2) A→S; B→Q; C→P; D→R

(3) A→P; B→P; C→S; D→R

(4) A→R; B→P; C→S; D→Q

36. The crystal field stabilization energy (CFSE) of $[Fe(H_2O)_6] Cl_2$ and $K_2[NiCl_4]$, respectively, are

(1) $-0.4\Delta_o$ and $-0.8\Delta_t$

(2) $-0.4\Delta_o$ and $-1.2\Delta_t$

(3) $-2.4\Delta_o$ and $-1.2\Delta_t$

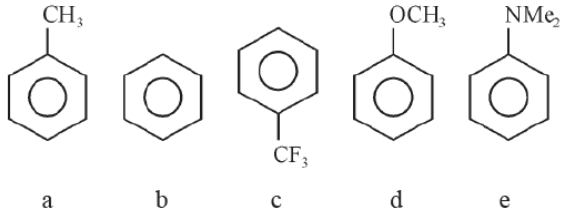
(4) $-0.6\Delta_o$ and $-0.8\Delta_t$

37. Among the given oxides of nitrogen; N_2O_3, N_2O_4 and N_2O_5, the compound/(s) having N–N bond is:

(1) N_2O_5 only (2) N_2O_4, N_2O_5 only

(3) N_2O_3, N_2O_4 only (4) N_2O_3 only

38. Decreasing order of reactivity towards electrophilic substitution for the following compounds is:



a b c d e

(1) c > b > a > d > e (2) e > d > a > b > c

(3) a > d > e > b > c (4) d > a > e > c > b

39. Compound A contains 8.7% Hydrogen, 74% Carbon and 17.3% Nitrogen. The molecular formula of the compound is, Given: Atomic masses of C, H and N are 12, 1 and 14 amu respectively. The molar mass of the compound A is 162 g mol^{-1}.

(1) $C_4H_6N_2$ (2) C_2H_3N

(3) C_5H_7N (4) $C_{10}H_{14}N_2$

40. In chromyl chloride, the number of d-electrons present on chromium is same as in (Given at no. of Ti : 22, V : 23, Cr : 24, Mn : 25, Fe : 26)

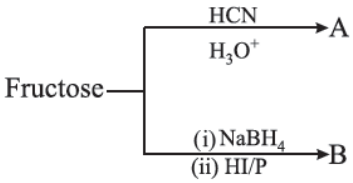
(1) Ti (III) (2) Fe (III)

(3) V (IV) (4) Mn (VII)

41. Staggered and eclipsed conformers of ethane are:
(1) Mirror images (2) Polymers
(3) Enantiomers (4) Rotamers

42. Which of the following is not a correct statement for primary aliphatic amines?
(1) The intermolecular association in primary amines is less than the intermolecular association in secondary amines.
(2) Primary amines can be prepared by the Gabriel phthalimide synthesis.
(3) Primary amines on treating with nitrous acid solution form corresponding alcohols except methyl amine.
(4) Primary amines are less basic than the secondary amines in gaseous phase

43. The formulas of A and B for the following reaction sequence are



(1) A = $C_7H_{14}O_8$, B = C_6H_{14}
(2) A = $C_7H_{13}O_7$, B = $C_7H_{14}O$
(3) A = $C_7H_{12}O_8$, B = C_6H_{14}
(4) A = $C_7H_{14}O_8$, B = $C_6H_{14}O_6$

44. In general, the properties that decreases and increase down a group in the periodic table respectively are
(1) Atomic radius and electrongegativity
(2) Electron gain enthalpy and electronegativity
(3) Electronegativity and atomic radius
(4) Electronegativity and electron gain enthalpy

45. From amongst the following alcohols the one that would react fastest with conc. HCl and anhydrous $ZnCl_2$, is
(1) 2–Butanol
(2) 2–Methylpropan–2–ol
(3) 2–Methylpropanol
(4) 1–Butanol

Integer Type Questions

46. A and B are two substances undergoing radioactive decay in a container. The half life of A is 15 min and that of B is 5 min. If the initial concentration of B is 4 times that of A and they both start decaying at the same time, how much time (in min) will it take for the concentration of both of them to be same?

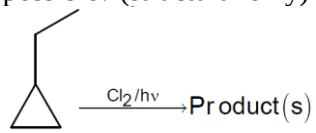
47. For the reaction $C_2H_6 \rightarrow C_2H_4 + H_2$, the reaction enthalpy Δ_rH =_____ kJ mol^{-1}. (Round off to the nearest integer).
[Given: Bond enthalpies in kJ mol^{-1} : C – C: 347, C = C: 611; C – H: 414, H – H: 436]

48. Consider the following reaction

$$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{+2} + 4H_2O, E^0 = 1.51V$$

The quantity of electricity required in Faraday to reduce five moles of MnO_4^- is_____
(Integer answer)

49. How many monochlorinated products (theoretically) of the following reaction is/are possible? (structural only)



50. For the reaction

$$A + B \rightleftharpoons 2C$$

The value of equilibrium constant is 100 at 298K. If the initial concentration of all the three species is 1M each, then the equilibrium concentration of C is $x \times 10^{-1}$M. The value of x is _____. (Nearest integer)

SECTION-III (MATHEMATICS)

Single Correct Type Questions

51. If α and β the roots of the equation $7x^2 - 3x - 2 = 0$, then the value of $\frac{\alpha}{1-\alpha^2} + \frac{\beta}{1-\beta^2}$ is equal to:

(1) $\frac{1}{24}$ (2) $\frac{27}{32}$
(3) $\frac{3}{8}$ (4) $\frac{27}{16}$

52. Let A be a 2 × 2 matrix with real entries such that $A^T = \alpha A + I$, where $\alpha \in \mathbb{R} - \{-1, 1\}$. If det $(A^2 - A) = 4$, then the sum of all possible real values of α is equal to
(1) 0
(2) $\frac{3}{2}$
(3) $\frac{5}{2}$
(4) 2

53. $2\sin\left(\frac{\pi}{22}\right)\sin\left(\frac{3\pi}{22}\right)\sin\left(\frac{5\pi}{22}\right)\sin\left(\frac{7\pi}{22}\right)\sin\left(\frac{9\pi}{22}\right)$ is equal to

(1) $\frac{3}{16}$ (2) $\frac{1}{16}$

(3) $\frac{1}{32}$ (4) $\frac{9}{32}$

54. Let $\left(-2-\frac{1}{3}i\right)^3 = \frac{x+iy}{27}(i=\sqrt{-1})$, where x and y are real number then $y - x$ equals:

(1) 91 (2) – 85

(3) 85 (4) – 91

55. In an isosceles triangle ABC, the vertex A is (6,1) and the equation of the base BC is $2x + y = 4$. Let the point B lie on the line $x + 3y = 7$. If (α, β) is the centroid of ΔABC, then $15(\alpha + \beta)$ is equal to:

(1) 39 (2) 41

(3) 51 (4) 63

56. One vertex of a rectangular parallelopiped is at the origin O and the lengths of its edges along x, y and z axes are 3, 4 and 5 units respectively. Let P be the vertex (3, 4, 5).

Then the shortest distance between the diagonal OP and an edge parallel to z axis, not passing through O or P is:

(1) $\frac{15}{\sqrt{5}}$ (2) $\frac{12}{5\sqrt{5}}$

(3) $12\sqrt{5}$ (4) $\frac{12}{5}$

57. The mean and the median of the following ten numbers in increasing order 10, 22, 26, 29, 34, x, 42, 67, 70, y are 42 and 35 respectively, then $\frac{y}{x}$ is equal to:

(1) $\frac{7}{3}$

(2) $\frac{9}{4}$

(3) $\frac{7}{2}$

(4) $\frac{8}{3}$

58. The series of positive multiples of 3 is divided into sets : {3}, {6, 9,12}, {15,18, 21, 24, 27}Then the sum of the elements in the 11th set is equal to

(1) 6912 (2) 7393

(3) 4999 (4) 6993

59. Let $x = 2t$, $y = \frac{t^2}{3}$ be a conic. Let S be the focus and B be the point on the axis of the conic such that $SA \perp BA$, where A is any point on the conic. If k is the ordinate of the centroid of ΔSAB, then $\lim\limits_{t\to 1} k$ is equal to

(1) $\frac{17}{18}$ (2) $\frac{19}{18}$

(3) $\frac{11}{18}$ (4) $\frac{13}{18}$

60. Let $f(x) = \begin{cases} \left|4x^2-8x+5\right|, \text{ if } 8x^2-6x+1 \geq 0 \\ \left[4x^2-8x+5\right], \text{ if } 8x^2-6x+1 < 0 \end{cases}$,

where $[\alpha]$ denotes the greatest integer less than or equal to α. Then the number of points in R where f is not differentiable is____.

(1) 1 (2) 2

(3) 3 (4) 4

61. Eight persons are to be transported from city A to city B in three cars of different makes. If each car can accommodate at most three persons, then the number of ways, in which they can be transported, is:

(1) 3360

(2) 1680

(3) 560

(4) 1120

62. The real function $f(x) = \frac{\text{cosec}^{-1}x}{\sqrt{x-[x]}}$, where $[x]$ denotes the greatest integer less than or equal to x, is defined for all x belonging to:

(1) All integers except 0, –1, 1

(2) All non-integers except the interval [–1, 1]

(3) All reals except the interval [–1, 1]

(4) All reals except integers

63. The integral $\int \frac{e^{3\log_e 2x} + 5e^{2\log_e 2x}}{e^{4\log_e x} + 5e^{3\log_e x} - 7e^{2\log_e x}}dx, x > 0$ is equal to:

(where c is a constant of integration)

(1) $\frac{1}{4}\log_e\left|x^2+5x-7\right| + c$

(2) $4\log_e\left|x^2+5x-7\right| + c$

(3) $\log_e\sqrt{x^2+5x-7} + c$

(4) $\log_e\left|x^2+5x-7\right| + c$

64. If the radius of the largest circle with centre (2, 0) inscribed in the ellipse $x^2 + 4y^2 = 36$ is r, then $12\,r^2$ is equal to

(1) 72 (2) 115

(3) 92 (4) 69

65. A quadratic equation, is chosen from the set of quadratic equations whose roots remains same on squaring the roots. The probability that selected quadratic equation has equal roots is

(1) 2/3 (2) 1/3

(3) 1/2 (4) 1

66. If the line $x - 1 = 0$, is a directrix of the hyperbola $kx^2 - y^2 = 6$, then the hyperbola passes through the point

(1) $(-2\sqrt{5}, 6)$ (2) $(-\sqrt{5}, 3)$

(3) $(\sqrt{5}, -2)$ (4) $(2\sqrt{5}, 3\sqrt{6})$

67. Let the solution curve of the differential equation $x\frac{dy}{dx} - y = \sqrt{y^2 + 16x^2}$, $y(1) = 3$ be $y = y(x)$. Then $y(2)$ is equal to:

(1) 15 (2) 11

(3) 13 (4) 17

68. The area (in sq. units) of the region, given by the set $\{(x, y) \in R \times R,\ x \geq 0,\ 2x^2 \leq y \leq 4 - 2x\}$ is:

(1) $\frac{17}{3}$ (2) $\frac{8}{3}$

(3) $\frac{7}{3}$ (4) $\frac{13}{3}$

69. A spherical iron ball of radius 10 cm is coated with a layer of ice of uniform thickness that melts at a rate of 50 cm^3/ min. When the thickness of the ice is 5 cm, then the rate at which the thickness (in cm/min) of the ice decreases, is:

(1) $\frac{1}{9\pi}$ (2) $\frac{5}{6\pi}$

(3) $\frac{1}{18\pi}$ (4) $\frac{1}{36\pi}$

70. Let $f(x) = 2 + |x| - |x-1| + |x+1|, x \in R$. Consider

$(S1): f'\left(-\frac{3}{2}\right) + f'\left(-\frac{1}{2}\right) + f'\left(\frac{1}{2}\right) + f'\left(\frac{3}{2}\right) = 2$

$(S2): \int_{-2}^{2} f(x)dx = 12$

Then,

(1) Both (S1) and (S2) are correct

(2) Only (S1) is correct

(3) Both (S1) and (S2) are wrong

(4) Only (S2) is correct

Integer Type Questions

71. The remainder when $(2023)^{2023}$ is divided by 35 is

72. If The circles $x^2 + y^2 + 6x + 8y + 16 = 0$ and $x^2 + y^2 + 2(3 - \sqrt{3})x + 2(4 - \sqrt{6})y = k + 6\sqrt{3} + 8\sqrt{6}, k > 0$, touch internally then k is equal to

73. Let $A = \{1, 2, 3, 4 \ldots\ldots 10\}$ and $B = \{0, 1, 2, 3, 4\}$. The number of elements in the relation $R = \{(a, b) \in A \times A: 2(a-b)^2 + 3(a-b) \in B\}$ is_____.

74. Let $\vec{a}$ and $\vec{b}$ be two vector such that $|\vec{a}| = \sqrt{14}, |\vec{b}| = \sqrt{6}$ and $|\vec{a} \times \vec{b}| = \sqrt{48}$. then $(\vec{a} \cdot \vec{b})^2$ is equal to _____.

75. The coefficient of x^4 in the expansion of $(1 + x + x^2 + x^3)^6$ in powers of x, is_____.



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